Supplementary Material

Variational Koopman Autoencoder Network with Mori–Zwanzig Closure for Stable and Interpretable Dynamics Learning

April 2, 2026

1 Burgers Equation

Table 1: Forecast MAE at last time step for the Burgers case.

Method	VKAN	FNO	DeepOnet	Transformer
Error	0.00061	0.00249	0.00357	0.00251

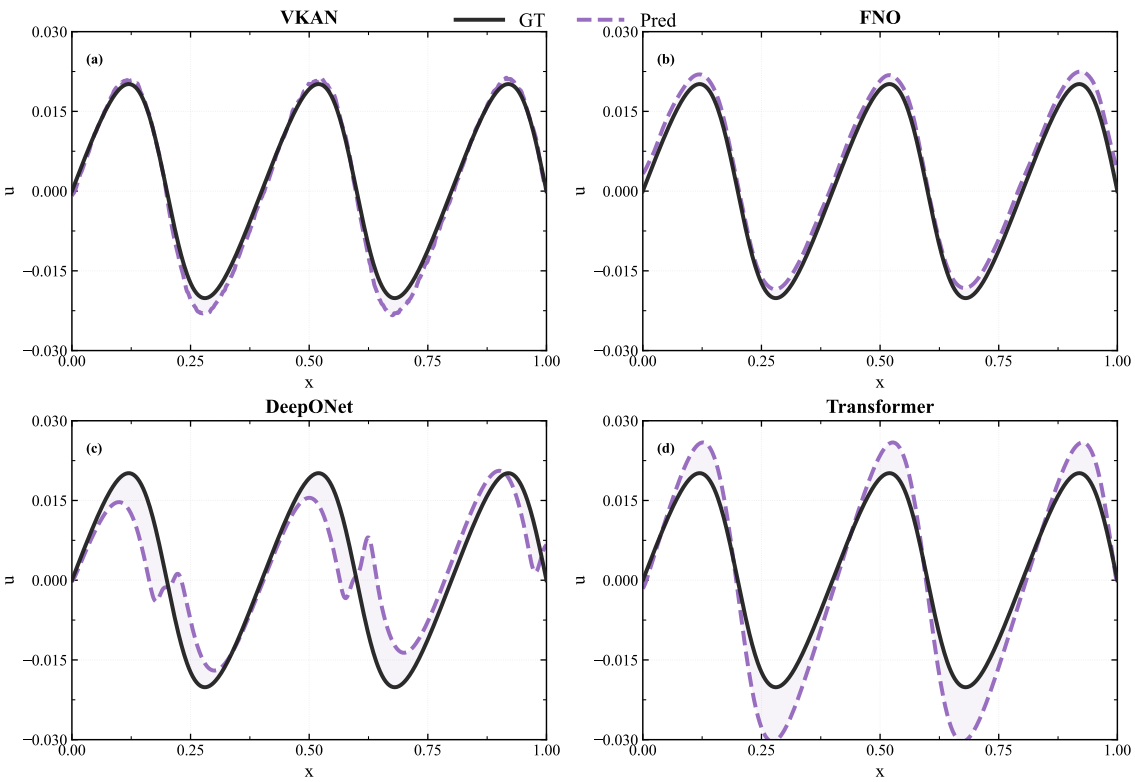

Figure 1: Qualitative result for the Burgers case.

2 Flow Past Cylinder

Table 2: Forecast MAE at last time step for the Cylinder flow.

Method	VKAN	Deepkoopman	CKAE	KoVAE
Error	0.0395	0.0686	0.0761	0.0736

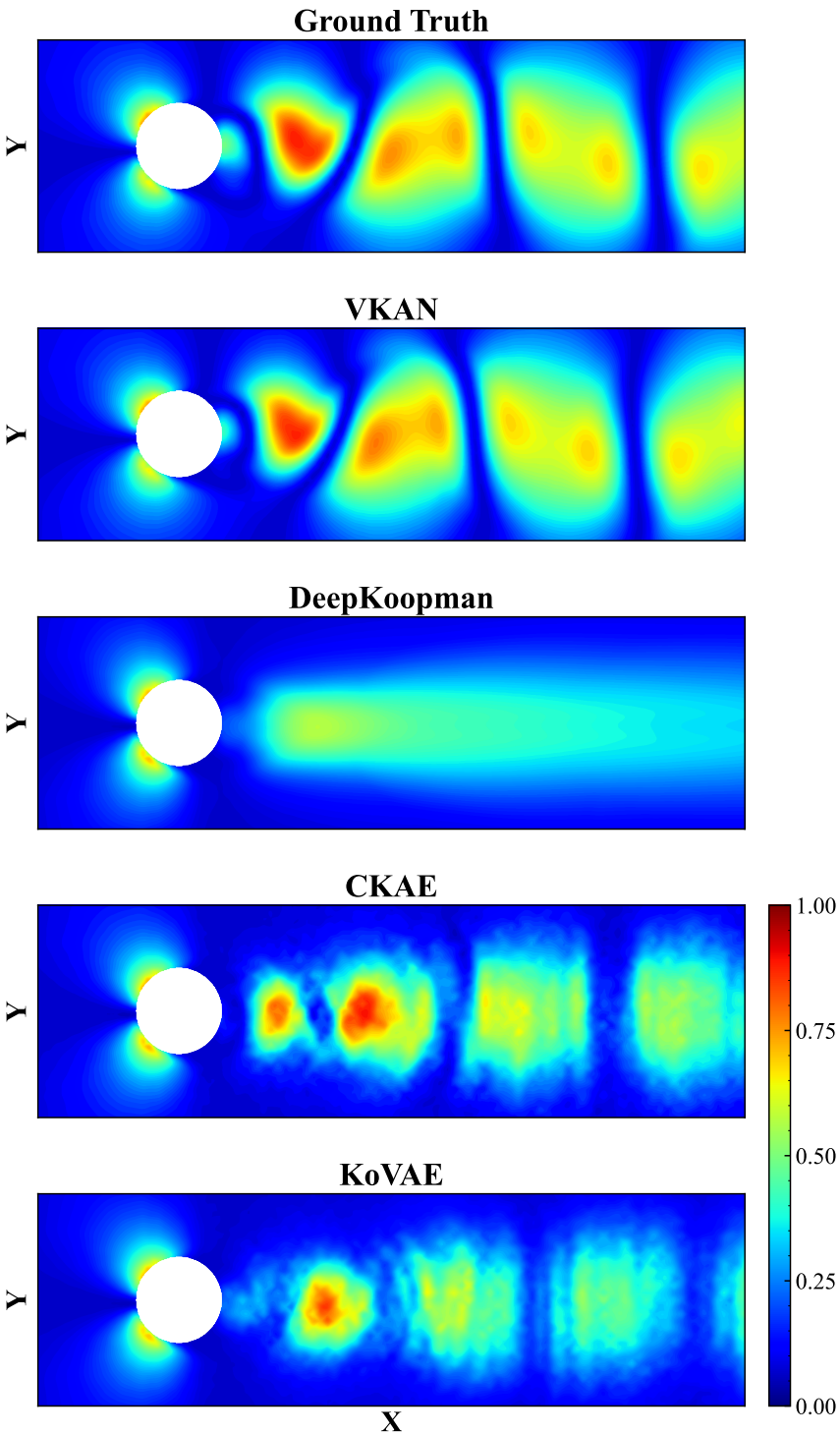

Figure 2: Qualitative result for the cylinder flow.

3 Kuramoto–Sivashinsky Equation

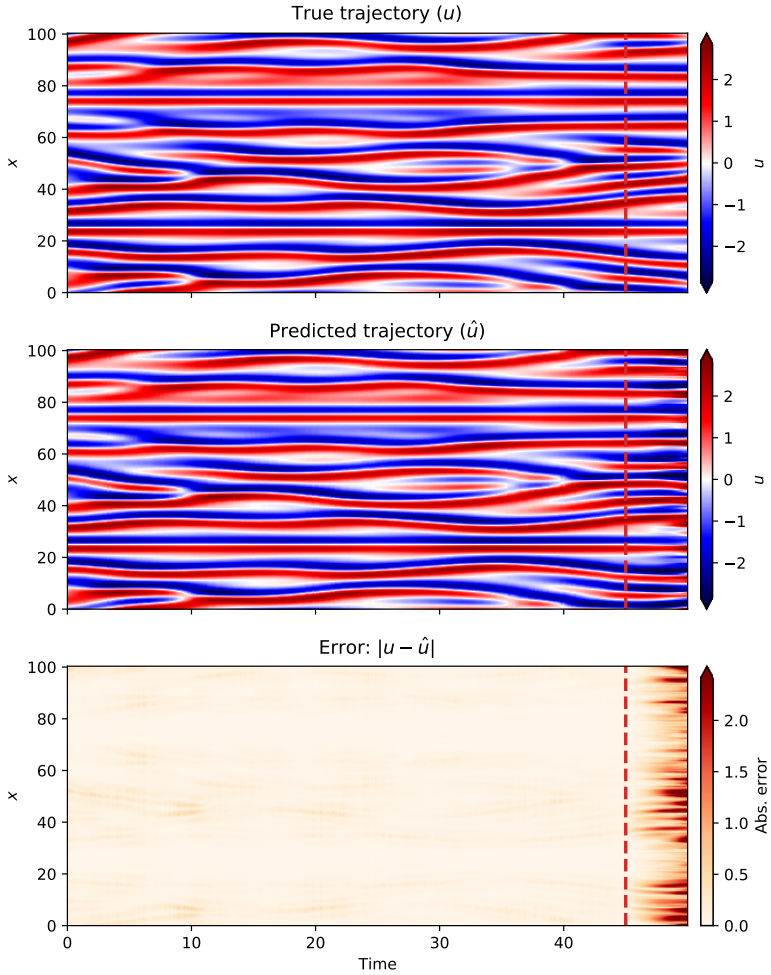

Figure 3: Qualitative result 1 for the Kuramoto–Sivashinsky equation.

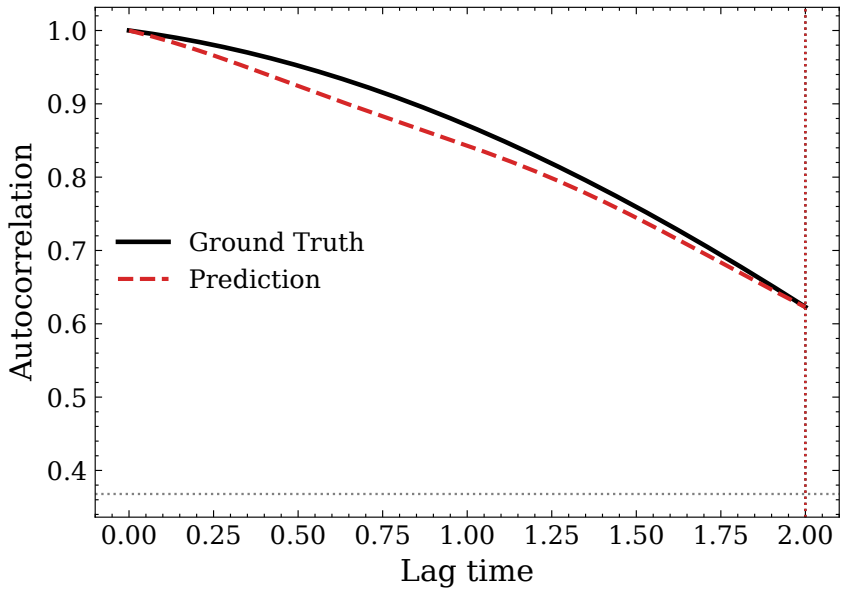

Figure 4: Qualitative result 2 for the Kuramoto–Sivashinsky equation.

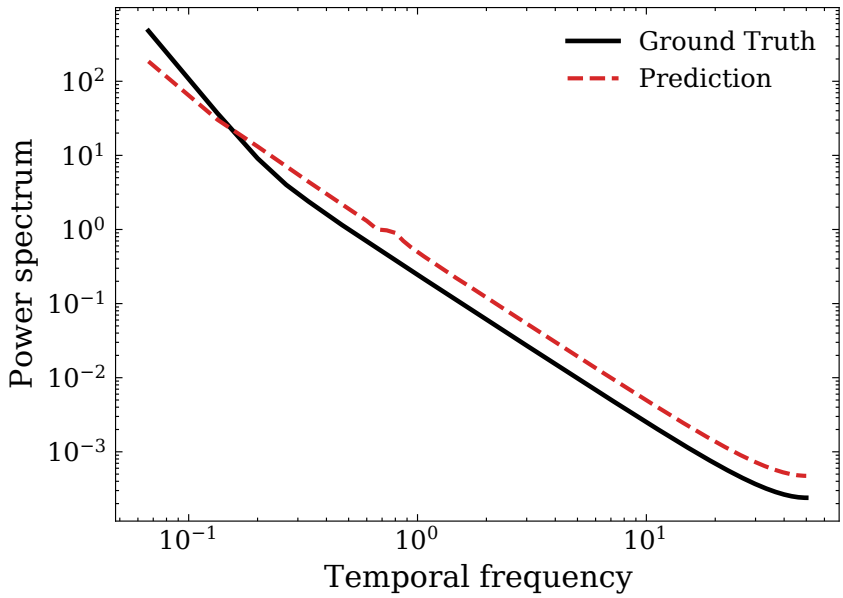

Figure 5: Qualitative result 3 for the Kuramoto–Sivashinsky equation.